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Practical No. 1

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1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 m=float(input())
2 v=float(input())
3 p=m*v
4 print(f"{p:.2f}kgm/s")
```

Average time: 0.006 s
8.78 ms

Maximum time: 0.007 s
7.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 6 ms

Expected output: 5.0
10.0
50.00kgm/s

Actual output: 5.0
10.0
50.00kgm/s

Test case 2 6 ms

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 n = int(input())
2 if 0<n<10:
3     print(f"{n**2}")
4 elif 10<=n<100:
5     print(f"{n**0.5:.2f}")
6 elif 100<=n<1000:
7     print(f"{n**(1/3):.2f}")
8 else:
9     print("Invalid")
```

Average time: 0.005 s
4.67 ms

Maximum time: 0.006 s
6.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 6 ms

Expected output: 9
81

Actual output: 9
81

Test case 2 6 ms

Test case 3 4 ms

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.
Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

1 from datetime import datetime
2
3
4
5 d1 = datetime.strptime(input(), "%Y-%m-%d")
6 d2 = datetime.strptime(input(), "%Y-%m-%d")
7
8 print((d2 - d1).days)

Average time: 0.010 s
10.25 ms

Maximum time: 0.027 s
27.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 27 ms

Expected output: 2014-07-02, 2014-11-02, 123

Actual output: 2014-07-02, 2014-11-02, 123

Test case 2 4 ms

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.
Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

1 number = int(input())
2 reverse_number = int(str(number)[::-1])
3 print(reverse_number)

Average time: 0.005 s
5.00 ms

Maximum time: 0.008 s
8.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 8 ms

Expected output: 5367, 7635

Actual output: 5367, 7635

Test case 2 4 ms

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1.1.5. Multiplication Table

100%

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
`<number> x <multiplier> = <result>`

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

Submit

Debugger

1n = int(input())
2for i in range(1,11):
3 print(f"{n} x {i} = {n*i}")

Average time0.004 s4.00 msMaximum time0.005 s5.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 15 msDebugTest cases

Expected outputActual output
88
8 x 1 = 88 x 1 = 8
8 x 2 = 168 x 2 = 16
8 x 3 = 248 x 3 = 24
8 x 4 = 328 x 4 = 32
8 x 5 = 408 x 5 = 40

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1.2.1. Pass or Fail

100%

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses

passorFa...

Submit

Debugger

1subject=int(input(""))
2marks=list(map(int,input("").split()))
3if any(mark<40 for mark in marks):
4 print("Fail")
5else:
6 total_marks=sum(marks)
7 Aggregate_percentage = (total_marks/(subject*100))*100
8 print(f"Aggregate Percentage:{Aggregate_percentage:.2f}")
9 if Aggregate_percentage > 75:
10 print("Grade: Distinction")
11 elif Aggregate_percentage > 60:

Average time0.005 s5.20 msMaximum time0.007 s7.00 ms5 out of 5 shown test case(s) passed5 out of 5 hidden test case(s) passed

Test case 17 msDebugTest cases

Expected outputActual output
44
55 65 78 8955 65 78 89
Aggregate Percentage: 71.75Aggregate Percentage: 71.75
Grade: First DivisionGrade: First Division

Test case 24 ms

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci....

```
1 # def fibonacci(n):
2
3 # return fibonacci()
4
5
6 n = int(input())
7 # for i in range(1, n + 1):
8 # print(fibonacci(i), end=" ")
9
10 def fib(n):
11     if n == 0:
```

Average time: 0.011 s 10.50 ms

Maximum time: 0.028 s 28.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 5 ms

Debug

Expected output: 0 1 1 2 3

Actual output: 0 1 1 2 3

Test case 2 4 ms

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Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

Submit

Debugger

1 num = int(input())
2 for i in range(1,num+1):
3 print("*" * i)

Average time: 0.004 s
Maximum time: 0.008 s
3.67 ms 8.00 ms

2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output: 5
*
* *
* * *
* * * *
* * * * *

Actual output: 5
*
* *
* * *
* * * *
* * * * *

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

Submit

Debugger

1 # Type Content here...
2 n = int(input())
3 for i in range(1,n+1):
4 for j in range(1,i+1):
5 print(j,end=' ')
6 print()
7

Average time: 0.004 s
Maximum time: 0.005 s
4.26 ms 5.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Actual output: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

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